

ROBOTICS (03/07/2019)

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The exam will be graded IFF the following recommendations have been taken into account:

- Write clearly so that the teacher can easily understand your answers
- Write your name, surname, and student id on each sheet you deliver for evaluation
- For each exercise/question report clearly the number and sub-number (if present)
- You are not allowed to use any programmable device (e.g., smartphone, calculator, etc.)
- You can use pen or pencil, paper will be provided, you cannot use notes or books

Exercise 1 – Sensors and Actuators [1+1+1+1+1 points]

Most mobile robots use electrical motors to actuate wheels and transform electrical power into torque.

- a) Compare DC motors and Servo motors
- b) What is a rotational encoder used for?
- c) How does an incremental rotational encoder work?
- d) How does an absolute rotational encoder work?
- e) What is the quadrature technique? In which kind of encoders is it used and why?

Exercise 2 – Robot Odometry [2+2+2 points]

The figure shows a few electric robots used for logistics (original picture of the Kiva System acquired by Amazon in 2012). It is a differential drive robot with the odometric center placed in the middle of the rectangular base. With reference to this kind of vehicle answer the following questions:

- a) What are direct and inverse kinematics equations used for in a differential drive robot?
- b) Compute the direct kinematics of the vehicle in the pictures and discuss the relevant parameters of such formulas.
- c) Control variables are v_r, v_l the right and left wheels velocities and $b = 0.4m$ the robot baseline. Assume the robot starts from position $[0, 0, 0]$ on the plane, where it will be after performing the following actions:
 - 1) For $0.2s$ keep $v_r = v_l = 3m/s$
 - 2) For $\frac{\pi}{10}s$ keep $v_r = -v_l = 1m/s$
 - 3) For $\frac{\pi}{5}s$ keep $v_r = 2 \times v_l = 2m/s$



Exercise 3 – Robot Local Path Planning [1+2+2]

Let's consider the problem of obstacle avoidance, a.k.a., local path planning, and the algorithms to realize it. Provide a description of

- a) The aim of local path planning and what should be its frequency with respect to global path planning
- b) Describe the Dynamic Window Approach (DWA) for local path planning
- c) DWA uses a navigation function, i.e., a function evaluating the value/cost of possible actions. Describe such navigation function and the parameters involved in it. Discuss briefly how it could be possible to set up such parameters in practice.

Exercise 4 – Robot Operating System [1 + 2 + 2 points]

In ROS you have **topics**, **services** and **actions**, (a) what is each of them used for, (b) what are the characteristics of each of them in terms of blocking/asynchronous behavior?

Exercise 5 – Robot Localization [1 + 1 + 2 + 2 points]

Let's consider the Monte Carlo Localization algorithm; with reference to a differential robot equipped with a planar lidar, i.e., 2D laser range finder, and a map in the form of an occupancy grid, answer the following:

- a) How does the Monte Carlo Localization algorithm work?
- b) What is the relationship between the Monte Carlo Localization algorithm and the recursive Bayes Filter used for localization?
- c) Describe the Beam Sensor Model in terms of its components and the rationale behind each of them
- d) How the Beam Sensor Model is used in Monte Carlo Localization considering an Occupancy grid as a map?